

LEGO[®] DIMENSIONS

Collection Tracker — Technical Documentation

Document	LEGO Dimensions Offline Collection Tracker
Version	v5 — June 2026
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Format	Single-file HTML + imgs/ folder — no server required
Sets	63 packs across Wave 1–9 + Wave 7.5
Lines	1176 total (HTML + CSS + JavaScript)

This document provides complete technical documentation for the LEGO Dimensions offline collection tracker — a single HTML file with no dependencies beyond a browser. It covers CSS architecture, HTML structure, JavaScript data model and logic, and maintenance instructions sufficient to fully reconstruct the tool from scratch.

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SECTION 1

Overview & Architecture

Design Principles

The tracker is a **fully self-contained single HTML file**. No server, no framework, no build step. All logic runs in the browser; state persists in **localStorage**. The only external dependency is Google Fonts — the page works without internet using system fonts.

File Structure

```
lego-dimensions/  
index.html <- Single file: HTML + CSS + JS (1176 lines)  
imgs/ <- Local images (63 .jpg files, white background)  
71171-starter.jpg  
71201-bttf-level.jpg  
... (61 more)  
download-helper.html <- Helper for manual image downloads
```

index.html Internal Layout

Section	Lines	Purpose
<head> / fonts	1–10	Charset, viewport, Google Fonts
<style> CSS	11–650	All visual styles — palette, components, animations
<header>	651–690	Portal SVG, title (Bangers font), stats bar
<div.controls>	691–730	Search input, 3 filter groups, toolbar
<div.sync-bar>	731–740	Export/Import buttons and toast
<div#output>	741	Dynamic render target — built by JS
<footer>	742–748	Legal disclaimer
<script> JS	749–1176	Data, filters, render, owned tracker, export/import

SECTION 2

CSS — Design System & Colour Palette

CSS Custom Properties (:root)

All colours and sizes are CSS custom properties. To retheme the entire app, only :root needs changing.

```
:root {
  --bg: #0a0a0a; /* Page background — near black */
  --surface: #1a1a1a; /* Card/panel background */
  --surface2: #222222; /* Alternate surface (table stripes) */
  --ld-yellow: #ffd700; /* Primary accent — LEGO Dimensions gold */
  --ld-cyan: #00b4d8; /* Secondary accent — portal blue */
  --ld-green: #00a651; /* Owned state / success */
  --ld-red: #e3000b; /* Level Pack badge */
  --ld-blue: #006cb7; /* Team Pack badge */
  --ld-white: #f5f5f0; /* Body text — warm white */
  --ld-muted: #555555; /* De-emphasised labels */
  --ld-muted2: #888888; /* Secondary muted text */
  --ld-border: rgba(255,215,0,0.12); /* Subtle yellow border */
  --ld-border2: rgba(255,215,0,0.30); /* Prominent yellow border */
  --radius: 8px; /* Default border radius */
  --card-w: 172px; /* Min card width in grid */
}
```

Pack Type Colours (JavaScript)

Type	Colour	Role
Starter Pack	#ffd700	Gold — the main entry set
Level Pack	#e3000b	Red — adventure/level packs
Team Pack	#006cb7	Blue — two-character packs
Fun Pack	#00b4d8	Cyan — single character packs
Story Pack	#a855f7	Purple — Year 2 story packs
Polybag	#f59e0b	Amber — limited/exclusive promotional sets

Key CSS Components

.card	Grid card — hover lift (translateY -5px), yellow top-border reveal
.card.owned	Green border + glow when item marked owned
.card.limited	Gold permanent border for EXCLUSIVE polybag
.card.exclusive	Purple permanent border for LIMITED EDITION polybag
.type-badge	Pack type label — top-left of card image, solid colour bg
.wave-badge	Wave label — bottom-left, dark semi-opaque bg (rgba 0,0,0,0.75)
.limit-badge	Star badge bottom-right for limited/exclusive cards
.owned-btn	Circular check button top-right of image
.placeholder	Shown when img fails to load — displays set number

<code>body::before</code>	Stud dot pattern background (28px grid radial-gradient)
<code>.portal-wrap div</code>	4 concentric rings, each counter-rotating at different speeds
<code>.sync-bar</code>	Export/Import toolbar below filters

SECTION 3

HTML Structure

Page Skeleton

```
<body>
<header>
<div class="header-bg"></div> <!-- dark bg + yellow line -->
<div class="portal-wrap"> <!-- 4 animated SVG rings -->
<svg class="ld-logo">...</svg> <!-- portal logo SVG -->
<div class="header-content">
<h1>LEGO <span>DIMENSIONS</span></h1>
<div class="stats"> <!-- Owned/Total/% counters -->
</div>
</header>

<div class="controls">
<input type="text" id="search"> <!-- live search -->
<div id="typeFilters"></div> <!-- built by buildFilters()-->
<div id="waveFilters"></div>
<div id="franchFilters"></div>
<div class="toolbar"> <!-- count + view toggle -->
</div>

<div class="sync-bar"> <!-- Export / Import -->
<button onclick="exportCollection()">
<input type="file" id="importFile">
<span id="syncToast"></span>
</div>

<div id="output"></div> <!-- JS renders cards here -->
<footer>...</footer>
<script>/* all logic */</script>
</body>
```

Dynamic Filter Buttons

The three filter groups (Type, Wave, Franchise) are built entirely by JS on page load via **buildFilters()**. Each button stores its value in a module variable (activeType / activeWave / activeFranch) and calls **render()** on click.

SECTION 4

Data Structure — Pack Catalogue

Pack Object Schema

Field	Type	Req	Description
num	string	Yes	LEGO set number — e.g. "71201"
name	string	Yes	Official name from lego-dimensions.fandom.com/wiki/
type	string	Yes	Starter Pack Level Pack Team Pack Fun Pack Story Pack Polybag
franchise	string	Yes	Franchise name — controls tag colour via franchColors map
wave	string	Yes	"Wave 1" through "Wave 9" plus "Wave 7.5"
figures	string	Yes	Comma-separated minifigures / platform note for Starter
img	string	Yes	Relative path: "imgs/NNNNN-description.jpg"
limited	string	No	If present: triggers limited/exclusive visual. Value = availability description.

Example Entries

```
{
  num:"71171", name:"Starter Pack",
  type:"Starter Pack", franchise:"LEGO", wave:"Wave 1",
  figures:"Batman, Gandalf, Wyldstyle · PS3/PS4/Xbox 360/Xbox One/Wii U",
  img:"imgs/71171-starter.jpg"
},
{
  num:"71342", name:"Green Arrow Polybag",
  type:"Polybag", franchise:"DC Comics", wave:"Wave 5",
  figures:"Green Arrow", img:"imgs/71342-dc-greenarrow.jpg",
  limited:"LIMITED EDITION · E3 2016 / GameStop Black Friday"
},
```

Adding a New Franchise Colour

```
/* In the franchColors object: */
"Indiana Jones": ["#8B4513", "#fff"],
/* Format: ["background hex", "text hex"] */
```

SECTION 5

JavaScript — Core Logic

Module State

Variable	Init	Purpose
activeType	"All"	Active pack type filter
activeWave	"All"	Active wave filter
activeFranch	"All"	Active franchise filter
viewMode	"grid"	"grid" or "list"
owned	Set([...])	Set of owned keys, from localStorage on load

Function Reference

buildFilters(id, items, getColor, getLabel, setActive)

Renders filter button group into DOM element #id. Calls setActive + render() on click.

makeCard(data, delay)

Builds one card DOM element. Handles img fallback, type/wave/limit badges, owned button, hover glow.

render()

Reads search + active filters, filters packs[], groups by waveOrder[], renders section labels and card grids into #output.

updateStats()

Updates #statOwned, #statTotal, #statPct in the header stats bar.

setView(mode)

Switches grid/list. Toggles .list-view class and re-renders.

itemKey(data)

Returns "num|name" string — unique key for localStorage.

exportCollection()

Serialises owned Set to JSON, triggers file download.

importCollection(event)

Reads JSON file, replaces owned Set, saves to localStorage, re-renders.

showToast(msg)

Shows status message in sync-bar for 3 seconds.

Filter Logic

```
const filtered = packs.filter(p => {
  if(activeType !== "All" && p.type !== activeType) return false;
  if(activeWave !== "All" && p.wave !== activeWave) return false;
  if(activeFranch !== "All" && p.franchise !== activeFranch) return false;
  if(q && !(
    p.name.toLowerCase().includes(q) ||
    p.num.includes(q) ||
    p.franchise.toLowerCase().includes(q) ||
    p.wave.toLowerCase().includes(q) ||
    p.type.toLowerCase().includes(q) ||
    (p.figures||"").toLowerCase().includes(q)
  )) return false;
```



```
return true;
});
/* Grouped render by waveOrder array – ensures correct wave sequence */
const waveOrder = ["Wave 1", "Wave 2", "Wave 3", "Wave 4", "Wave 5",
"Wave 6", "Wave 7", "Wave 7.5", "Wave 8", "Wave 9"];
```

Owned State Persistence

```
/* Load on page init */
const owned = new Set(JSON.parse(
localStorage.getItem("ld_owned") || "[]"
));
/* Toggle on card button click */
if(owned.has(key)) { owned.delete(key); }
else { owned.add(key); }
localStorage.setItem("ld_owned", JSON.stringify([...owned]));
updateStats();
```

SECTION 6

Export / Import — Collection Sync

Allows backing up and transferring the collection between devices with no server. Uses the Blob URL download trick and the browser File API.

Export

```
function exportCollection(){
const data = {
version: 1,
exported: new Date().toISOString(),
owned: [...owned] // Set -> Array for JSON serialisation
};
const blob = new Blob([JSON.stringify(data, null, 2)],
{type:"application/json"});
const url = URL.createObjectURL(blob);
const a = document.createElement("a");
a.href = url;
a.download = "lego-dimensions-collection.json";
a.click();
URL.revokeObjectURL(url); // free memory immediately
showToast("Collection exported!");
}
```

Import

```
function importCollection(event){
const file = event.target.files[0];
if(!file) return;
const reader = new FileReader();
reader.onload = e => {
try {
const data = JSON.parse(e.target.result);
const keys = Array.isArray(data.owned) ? data.owned : [];
if(!keys.length){ showToast("No data found."); return; }
owned.clear();
keys.forEach(k => owned.add(k));
localStorage.setItem("ld_owned", JSON.stringify([...owned]));
render();
showToast(`Imported ${keys.length} owned items!`);
} catch(err){ showToast("Invalid file."); }
event.target.value = ""; // allow re-import of same file
};
reader.readAsText(file);
}
```

JSON Backup File Format

```
{
"version": 1,
"exported": "2026-06-08T14:30:00.000Z",
"owned": [
"71171|Starter Pack",
"71201|Back to the Future Level Pack",
"71340|Supergirl Polybag"
]
}
/* Key format: "num|name" — human-readable and unique */
```

SECTION 7

Limited / Exclusive Logic

The optional **limited** field triggers special visual treatment. Two sets use this — both polybags. Note: no LEGO Dimensions NFTs exist.

Set	Name	Label on Pack	How to Obtain
71340	Supergirl Polybag	EXCLUSIVE	Included with PS4 Year 2 Starter Pack relaunch (2016)
71342	Green Arrow Polybag	LIMITED EDITION	E3 2016 (convention) · GameStop Black Friday 2016 · Preorder bonus B

makeCard() logic for limited field:

```

if(data.limited){
const isExcl = data.limited.startsWith('EXCLUSIVE');
// CSS class: .exclusive (purple border) or .limited (gold border)
card.classList.add(isExcl ? 'exclusive' : 'limited');
// Star badge bottom-right of card image
const lb = document.createElement('div');
lb.className = 'limit-badge ' + (isExcl ? 'excl' : 'ltd');
lb.textContent = isExcl ? '* Exclusive' : '* Limited Edition';
lb.title = data.limited; // full text on hover
imgA.appendChild(lb);
// Availability line in card body
const info = document.createElement('div');
info.style.color = isExcl ? '#a855f7' : '#fbbf24';
info.textContent = data.limited;
body.appendChild(info);
}

```

SECTION 8

Full Pack Catalogue

All 63 sets — verified against lego-dimensions.fandom.com/wiki/

Wave 1 — 21 packs

#	Set	Name	Type	Franchise
1	71171	Starter Pack	Starter Pack	LEGO
2	71201	Back to the Future Level Pack	Level Pack	Back to the Future
3	71202	The Simpsons Level Pack	Level Pack	The Simpsons
4	71203	Portal 2 Level Pack	Level Pack	Portal 2
5	71205	Jurassic World Team Pack	Team Pack	Jurassic World
6	71206	Scooby-Doo! Team Pack	Team Pack	Scooby-Doo!
7	71209	Wonder Woman Fun Pack	Fun Pack	DC Comics
8	71210	Cyborg Fun Pack	Fun Pack	DC Comics
9	71212	Emmet Fun Pack	Fun Pack	The LEGO Movie
10	71213	Bad Cop Fun Pack	Fun Pack	The LEGO Movie
11	71214	Benny Fun Pack	Fun Pack	The LEGO Movie
12	71215	Jay Fun Pack	Fun Pack	Ninjago
13	71216	Nya Fun Pack	Fun Pack	Ninjago
14	71217	Zane Fun Pack	Fun Pack	Ninjago
15	71218	Gollum Fun Pack	Fun Pack	The Lord of the Rings
16	71219	Legolas Fun Pack	Fun Pack	The Lord of the Rings
17	71220	Gimli Fun Pack	Fun Pack	The Lord of the Rings
18	71221	Wicked Witch Fun Pack	Fun Pack	The Wizard of Oz
19	71222	Laval Fun Pack	Fun Pack	Legends of Chima
20	71223	Cragger Fun Pack	Fun Pack	Legends of Chima
21	71232	Eris Fun Pack	Fun Pack	Legends of Chima

Wave 2 — 5 packs

#	Set	Name	Type	Franchise
1	71204	Doctor Who Level Pack	Level Pack	Doctor Who
2	71207	Ninjago Team Pack	Team Pack	Ninjago
3	71211	Bart Simpson Fun Pack	Fun Pack	The Simpsons
4	71227	Krusty the Clown Fun Pack	Fun Pack	The Simpsons
5	71231	Unikitty Fun Pack	Fun Pack	The LEGO Movie

Wave 3 — 5 packs

#	Set	Name	Type	Franchise
1	71228	Ghostbusters Level Pack	Level Pack	Ghostbusters
2	71229	DC Comics Team Pack	Team Pack	DC Comics
3	71230	Doc Brown Fun Pack	Fun Pack	Back to the Future
4	71234	Sensei Wu Fun Pack	Fun Pack	Ninjago
5	71238	Cyberman Fun Pack	Fun Pack	Doctor Who

Wave 4 — 7 packs

#	Set	Name	Type	Franchise
1	71233	Stay Puft Fun Pack	Fun Pack	Ghostbusters
2	71235	Midway Arcade Level Pack	Level Pack	Midway Arcade
3	71236	Superman Fun Pack	Fun Pack	DC Comics
4	71237	Aquaman Fun Pack	Fun Pack	DC Comics
5	71239	Lloyd Fun Pack	Fun Pack	Ninjago
6	71240	Bane Fun Pack	Fun Pack	DC Comics
7	71241	Slimer Fun Pack	Fun Pack	Ghostbusters

Wave 5 — 2 packs

#	Set	Name	Type	Franchise
1	71340	Supergirl Polybag	Polybag	DC Comics
2	71342	Green Arrow Polybag	Polybag	DC Comics

Wave 6 — 7 packs

#	Set	Name	Type	Franchise
1	71242	Ghostbusters (2016) Story Pack	Story Pack	Ghostbusters 2016
2	71244	Sonic the Hedgehog Level Pack	Level Pack	Sonic the Hedgehog
3	71245	Adventure Time Level Pack	Level Pack	Adventure Time
4	71246	Adventure Time Team Pack	Team Pack	Adventure Time
5	71247	Harry Potter Team Pack	Team Pack	Harry Potter
6	71248	Mission: Impossible Level Pack	Level Pack	Mission: Impossible
7	71251	The A-Team Fun Pack	Fun Pack	The A-Team

Wave 7 — 4 packs

#	Set	Name	Type	Franchise
1	71253	Fantastic Beasts Story Pack	Story Pack	Fantastic Beasts
2	71256	Gremlins Team Pack	Team Pack	Gremlins
3	71257	Tina Goldstein Fun Pack	Fun Pack	Fantastic Beasts

4	71258	E.T. the Extra-Terrestrial Fun Pack	Fun Pack	E.T.
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Wave 7.5 — 4 packs

#	Set	Name	Type	Franchise
1	71264	The LEGO Batman Movie Story Pack	Story Pack	The LEGO Batman Movie
2	71285	Marceline Fun Pack	Fun Pack	Adventure Time
3	71286	Michael Knight Fun Pack	Fun Pack	Knight Rider
4	71344	Excalibur Batman Fun Pack	Fun Pack	The LEGO Batman Movie

Wave 8 — 3 packs

#	Set	Name	Type	Franchise
1	71266	Chase McCain Fun Pack	Fun Pack	LEGO City
2	71267	The Goonies Level Pack	Level Pack	The Goonies
3	71348	Hermione Granger Fun Pack	Fun Pack	Harry Potter

Wave 9 — 5 packs

#	Set	Name	Type	Franchise
1	71255	Teen Titans Go! Team Pack	Team Pack	Teen Titans Go!
2	71287	Starfire Fun Pack	Fun Pack	Teen Titans Go!
3	71343	Buttercup Fun Pack	Fun Pack	The Powerpuff Girls
4	71346	The Powerpuff Girls Team Pack	Team Pack	The Powerpuff Girls
5	71349	Beetlejuice Fun Pack	Fun Pack	Beetlejuice

SECTION 9

Maintenance Guide

Adding a New Set

1. Download official white-background figure photo from cdn.rebrickable.com/media/sets/NNNNN-1.jpg
2. Save as **NNNNN-description.jpg** in the **imgs/** folder
3. Open **index.html** in any text editor
4. Locate **const packs = [...]** in the **<script>** block
5. Add new object in the correct wave section (follow existing format)
6. Save and reload — no build step, no server needed

New Set Template

```
{  
  num:"NNNNN",  
  name:"Character Name Type Pack",  
  type:"Fun Pack",  
  franchise:"Franchise Name",  
  wave:"Wave N",  
  figures:"Character Name",  
  img:"imgs/NNNNN-description.jpg"  
},
```

Deploying to GitHub Pages

1. Create GitHub repo (e.g. **lego-dimensions-tracker**)
2. Push **index.html** + **imgs/** folder to the **main** branch
3. Settings → Pages → Source: main branch → / (root) → Save
4. Live at <https://username.github.io/lego-dimensions-tracker/>
5. Google Fonts requires internet; all other functionality is fully offline

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Fan-made tool, no commercial purpose.